



AI Action Plan  
Attn: Faisal D'Souza  
National Coordination Office  
2415 Eisenhower Avenue  
Alexandria, VA 22314

Re: Memo to the White House on AI Action Plan

**Overview: AI Infrastructure**

To maintain the United States' competitive edge in artificial intelligence, it is important to invest in the foundational infrastructure that powers AI development: advanced semiconductor manufacturing and materials for next-generation data centers. Traditional data center architectures are straining under the exponential growth of AI workloads, requiring a shift toward novel materials and cooling technologies that enhance energy efficiency and computational capacity. The U.S. must lead in developing new materials and autonomous manufacturing techniques that will enable AI systems to operate faster and more efficiently while optimizing resource use and impact. Expanding U.S. semiconductor fabrication capabilities is critical, not only to reduce reliance on foreign supply chains, but also to drive innovation in chip designs tailored for AI-specific applications to ensure the U.S. can position itself as the premiere global hub for next-generation AI infrastructure.

Furthermore, the next frontier in semiconductor manufacturing lies beyond Earth's atmosphere. Space-based manufacturing presents an opportunity to produce higher-purity materials and essentially defect-free chips by leveraging the ultra-high vacuum of space, unlocking breakthroughs in AI hardware performance. By utilizing high-cadence, reusable commercial space capabilities, U.S. industry can establish orbital semiconductor production facilities that push the limits of chip efficiency and durability. A national AI infrastructure strategy must prioritize investments in both terrestrial and space-based semiconductor manufacturing - an area the People's Republic of China is already trying to gain advantage in - ensuring the U.S. remains a leader across the entire value chain of AI-driven innovations to promote economic competitiveness and underpin U.S. national security.



### **WH AI Action Plan Recommendations:**

#### Advance AI-Specific Semiconductor Manufacturing in the U.S.

- Expand government funding to support R&D in manufacturing next-generation semiconductor materials with inherently higher thermal conductivity.

#### Invest in Next-Generation AI Data Centers

- Fund research into alternative materials for AI training and inferencing workloads.
- Develop national AI data center standards to improve efficiency, optimize resource management, and enhance cybersecurity.
- Partner with leading technology firms and energy companies to create durable, high-performance AI data centers that integrate flexible energy sources.

#### Expand U.S. Leadership in Space-Based Semiconductor Manufacturing

- Launch a national initiative to support orbital semiconductor manufacturing through the creation of a coalition among private space companies, data center operators, chip makers, and SIA members.
- Fund pilot programs for orbital fabrication of AI chips, testing their performance and scalability for commercial production.

#### Secure AI Supply Chains

- Strengthen U.S. semiconductor supply chains through strategic stockpiling of critical raw materials and rare earth elements.
- Recognize space-based semiconductor manufacturing capabilities are crucial to strengthening the US Industrial Base, which includes the production of microelectronic materials essential for advanced AI systems.

On behalf of the Besxar team, we are inspired by the White House leadership in developing an AI Action Plan and look forward to contributing however we can to ensure American leadership.

#### **Submitted by:**

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